T3 Weinberg-operator coefficient comparison

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \tilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\text{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\text{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\text{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$

$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$

Model	Ordered coefficient extracted from Matchete	Symmetric physical Wilson coefficient
T3-E	$A_5^{ij} =$	$C_5^{ij} =$

$$-\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}y_{2,j,r}}m_2^2\lambda_5 m_1^4}{\epsilon M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}\right.$$

$$\frac{2}{3}\left(\frac{\sqrt{6}\hbar\overline{y_{1,i,r}}\sqrt{6}\hbar\overline{y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}+\right.$$

$$+\frac{\sqrt{6}\hbar\overline{y_{1,i,r}}\sqrt{6}\hbar\overline{y_{2,j,r}}m_2^2\lambda_5 m_1^4}{M_F\left(\log\left(\frac{M_F^2}{m_1^2}\right)M_F^2m_1^2-\log\left(\frac{M_F^2}{m_1^2}\right)m_2^2m_1^2+\log\left(\frac{M_F^2}{m_2^2}\right)m_2^2m_1^2-\log\left(\frac{M_F^2}{m_2^2}\right)M_F^2m_2^2\right)}$$